MASTER FINAL PROJECT

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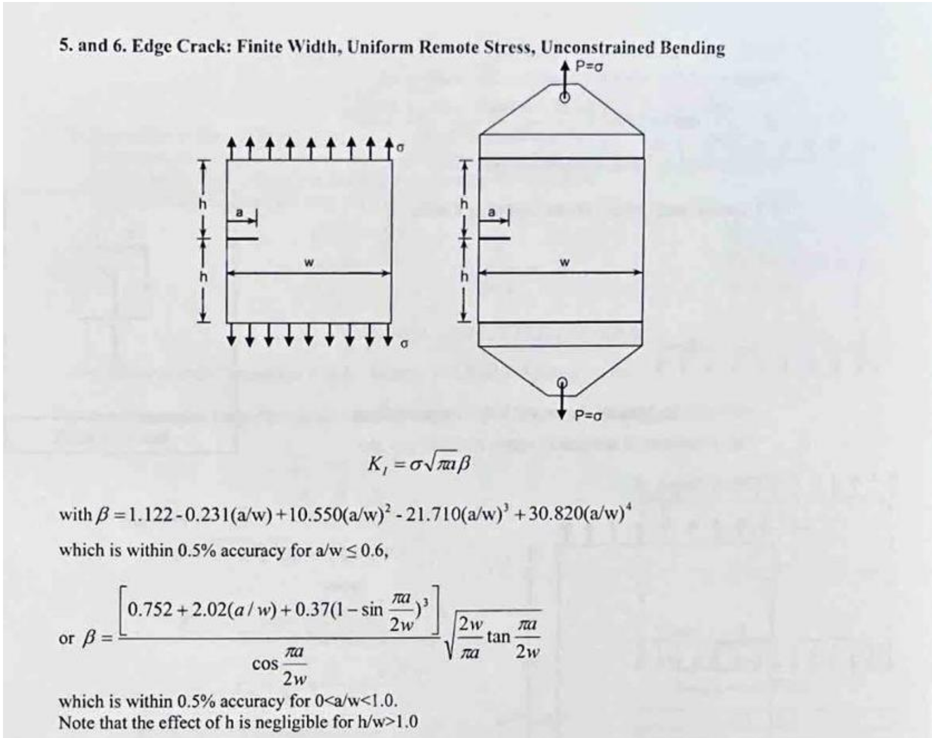
1 Abstract

Detailed finite element models(DFEMS) are quickly becoming the new standard for all new development of aerospace structures. However, as refined as the DFEMs are becoming, they still tend to be highly refined loads models and not true stress models. This forces analysts to revert back to using simple kt charts from resources like Peterson. This is often overly conservative with geometry that is not truly reflective of the designed geometry. For a flat plate model containing compounding Kts, this paper will develop a python based finite element solver that will create a "break out" FEM model, re-meshing critical locations with refined 8 noded quadrilateral elements, while leaving the surrounding non-critical elements as 4 noded quadrilateral elements. To bridge the higher and lower order element mesh, 7 noded transition elements are developed and implemented. Freebody loads are then applied from the DFEM to get true stress concentration values that can be used for classical Linear Elastic Fracture Mechanics analysis of a corner cracked plate.

Results of the refined mixed mesh plate will be compared against classical elasticity solutions to ensure validity. Additionally, results will be compared against traditional element refinement methods relying on a substantial increase in lower order elements to determine the performance differences between the two refinement methods.

2 Problem Statement

Within the industry of Fatigue and Damage Tolerance (F&DT) engineering for aviation, Damage Tolerance (DT) analysis is done using commercially available Linear Elastic Fracture Mechanics (LEFM) like NASGROW or AFGROW. Figure 1 shows a typical through thickness edge crack for a finite width plate subject to tension load. In this solution, beta accounts for the changes in stress intensity as the crack grows along the width of the plate.



5. and 6. Edge Crack: Finite Width, Uniform Remote Stress, Unconstrained Bending

$$K_I = \sigma\sqrt{\pi a}\,\beta$$

with $\beta = 1.122 - 0.231(a/w) + 10.550(a/w)^2 - 21.710(a/w)^3 + 30.820(a/w)^4$

which is within 0.5% accuracy for $a/w \leq 0.6$,

$$\text{or } \beta = \frac{\left[0.752 + 2.02(a/w) + 0.37(1 - \sin\frac{\pi a}{2w})^3\right]}{\cos\frac{\pi a}{2w}}\sqrt{\frac{2w}{\pi a}\tan\frac{\pi a}{2w}}$$

which is within 0.5% accuracy for $0 < a/w < 1.0$.
Note that the effect of h is negligible for $h/w > 1.0$

Figure 1: Example Through Thickness Edge Crack Finite Width Plate Solution [1]

However, in actual aerostructures, a pure finite width plate is never actually seen. Thus, it is necessary to make assumptions between actual geometry and the solution's assumed geometry. For example, a representative vertical frame is shown in Figure 2, and Figure 3 shows the idealized structure that would match the solutions assumed geometry.

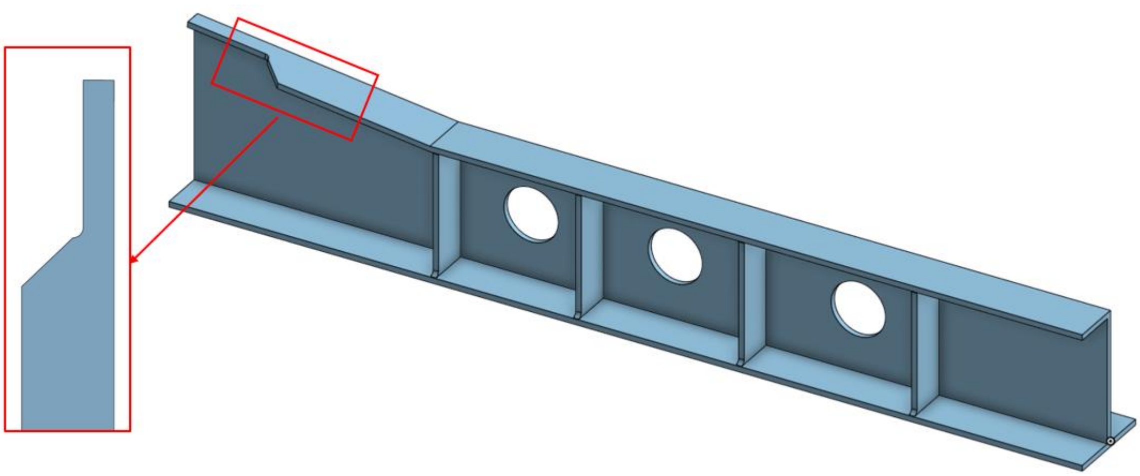


Figure 2: Representative Vertical Frame

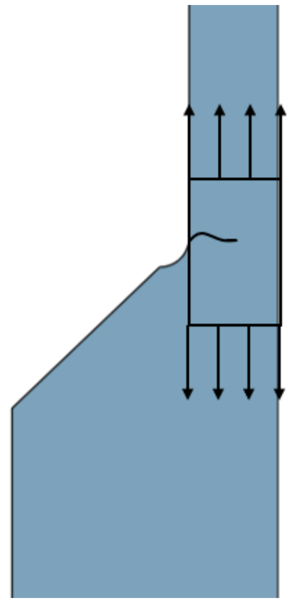


Figure 3: Assumed Finite Width Through Thickness Edge Crack Specimen

However, with the actual geometry reducing plate widths and running out into a fillet, a stress concentration will be present. The through thickness edge crack for a finite width plate has no means of capturing this stress concentration thus a Kt must be calculated and applied on top of the traditional solution's beta factor. This changes the stress intensity calculation to be as shown below.

$$K = \sigma K t \beta * \sqrt{\pi a}$$

Figure 4 shows a typical AVFEM or loads DFEM mesh for detail shown in Figure 3. This mesh is insufficient to capture the stress concentration effects. Thus industry common practice is to use the model to far-field loads and apply a Kt. Traditionally this Kt value would be calculated from an industry standard handbook like Peterson's Stress Concentrations [2] (Figure 6). However, again the models within [2] do not perfectly match the actual structure. Leading to analyst having to choose between possibly overly conservative or possibly un-conservative Kt value as shown in Figure 5. Depending on the effect of the additional plate width, the actual Kt could range from 1.82 to 2.6 for the geometry shown in Figure 5.

With this subprime mesh shown and the limitations of LEFM solutions and Kt solutions explained, the goal of this paper is to develop a finite element solver within python that is capable of solving standard Nastran elements like: 3 node triangles, 4 node quadrilaterals, and 8 node quadrilaterals. Additionally, the solver will be capable of solving transition elements that are 6 or 7 node quadrilaterals. These transition elements bridge the gap between 4 node and 8 node quadrilaterals.

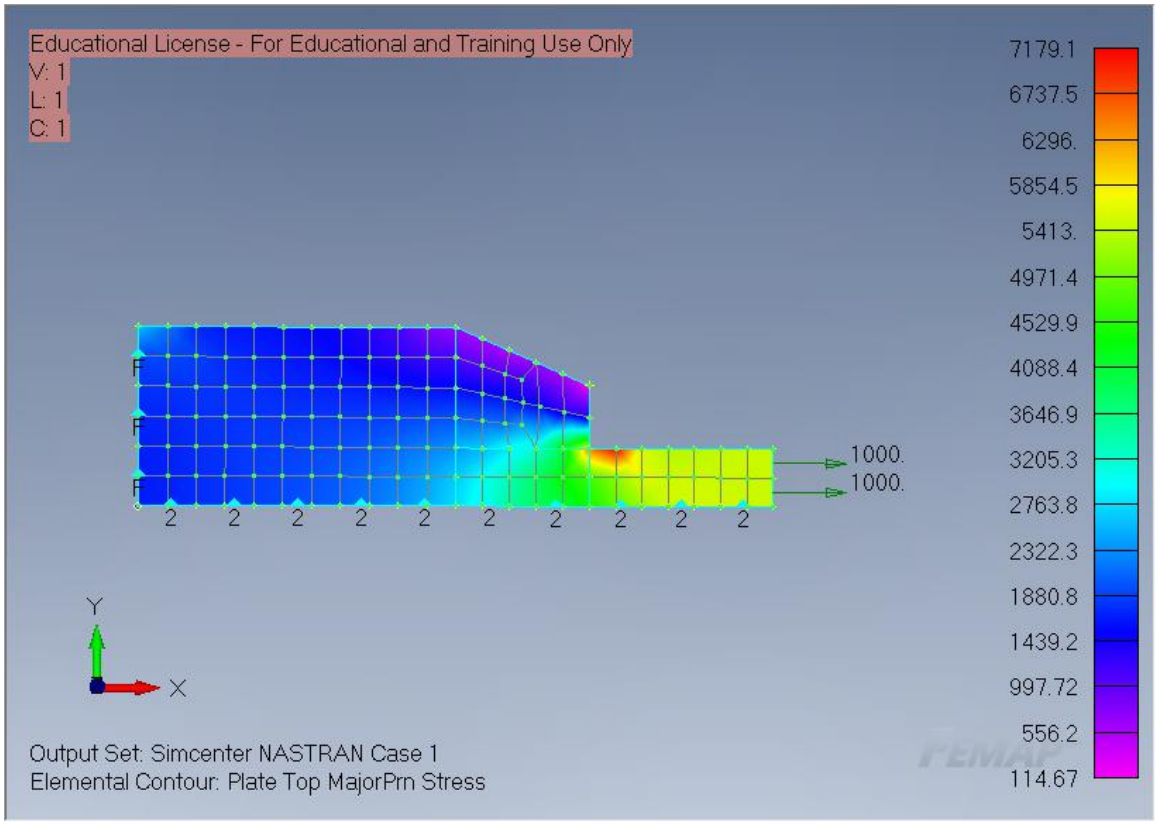


Figure 4: AVFEM Representation

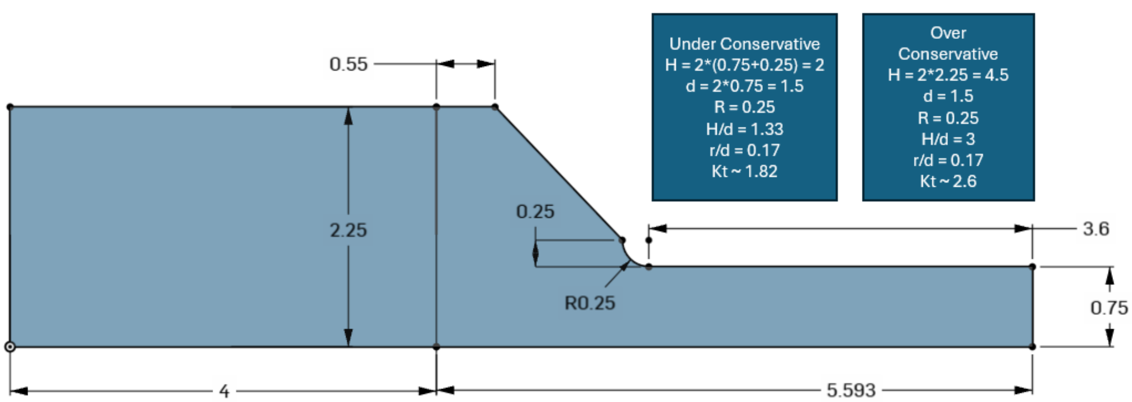


Figure 5: Specimen Geometry

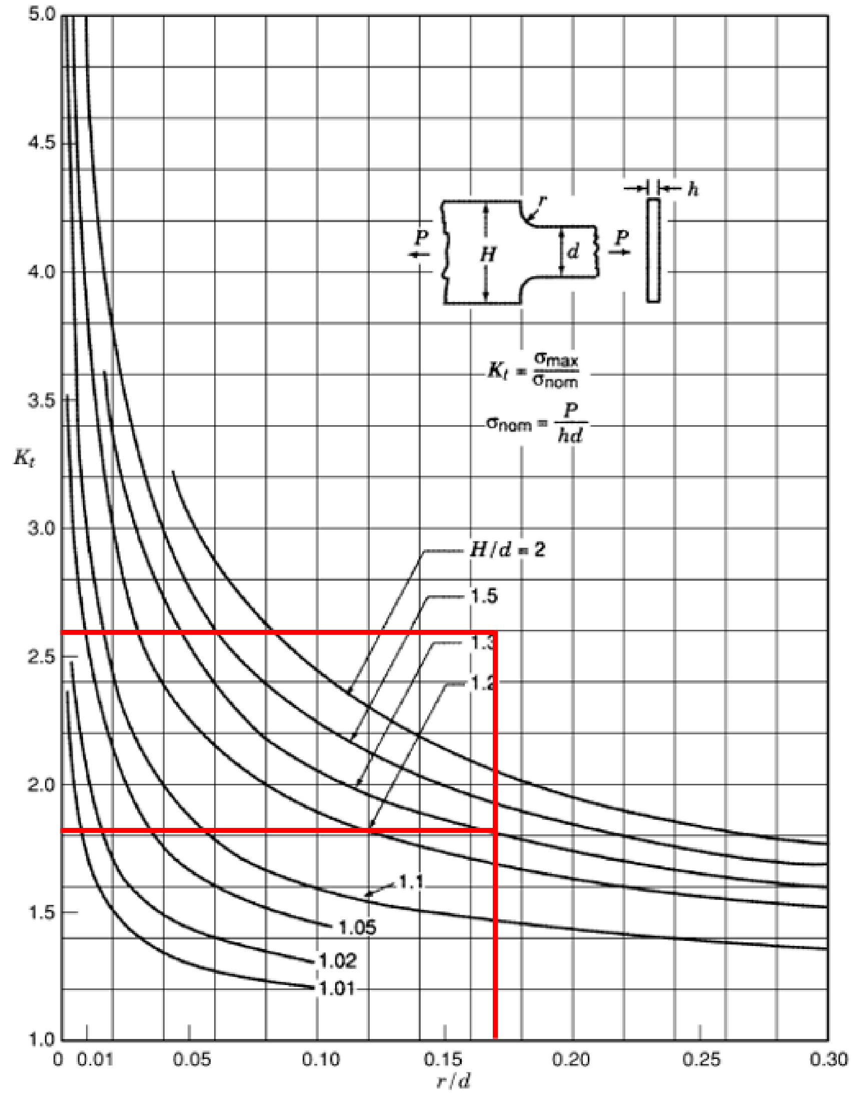


Chart 3.1 Stress concentration factors K_t for a stepped flat tension bar with shoulder fillets (based on data of Frocht 1935; Appl and Koerner 1969; Wilson and White 1973).

Figure 6: Chart 3.1 Petersons Stress Concentrations [2]

3 Finite Element Method Derivation

The four parts of the structural problem.

a) Constitutive
b) Kinematics
c) Equilibrium
d) Boundary Conditions

3.1 Constitutive Equations

The constitutive equations relate stress to strain using hookes law. Two assumptions can be made that either the element is in plane strain ($\epsilon_{zz} = 0$) or plane stress ($\sigma_{zz} = 0$). Both relations are given below. As the assumed thickness throughout all plates, plane stress is used in all analysis represented by matrix D.

Stress vector:

$$\{\sigma\}_{3d} = [\sigma_x \quad \sigma_y \quad \sigma_z \quad \tau_{xy} \quad \tau_{yz} \quad \tau_{zx}]^T$$

$$\{\sigma\}_{2d} = [\sigma_x \quad \sigma_y \quad \tau_{xy}]$$

Strain vector:

$$\{\epsilon\}_{3d} = [\epsilon_x \quad \epsilon_y \quad \epsilon_z \quad \gamma_{xy} \quad \gamma_{yz} \quad \gamma_{zx}]^T$$

$$\{\epsilon\}_{2d} = [\epsilon_x \quad \epsilon_y \quad \epsilon_{xy}]$$

Constitutive Equation:

$$\{\sigma\} = [D]\{\epsilon\}$$

The plane strain relation is defined as the following for the 2 dimensional assumption. [3]

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ -\nu & 1 & 0 \\ 0 & 0 & \dfrac{1-2\nu}{2} \end{bmatrix} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{Bmatrix}$$

The plane stress relation is defined as the following can be used for the 2d assumption. [3]

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{Bmatrix}$$

3.2 Kinematic Equations, Equilibrium

For a liner-elastic body the total potential energy statement is shown below.

$$\Pi(u) = U_I(u) - U_E(u)$$

U_I is the internal potential energy (strain energy) and U_E is the external potential energy in the form of external forces and u is the global displacement field.

The internal potential energy is defined as strain energy. This is the work done by the internal stresses while deforming a body from zero strain to its final strain state.

$$dU_I = \sigma^T d\epsilon$$

Integrating to get the total potential energy

$$U_I = \int_V \int_0^{\epsilon} \sigma^T \, d\epsilon$$

Substituting the constitutive matrix $\sigma = D\epsilon$

$$U = \int_V \int_0^{\epsilon} (D\epsilon)^T d\epsilon$$

Integrating with respect to strain yields

$$U_I(u) = \frac{1}{2}\int_V \epsilon^T D\epsilon dV$$

External Work

Assuming there are no body forces or surface tractions

$$U_E(u) = \sum_i u_i^T P_i$$

This gives the potential energy statement of:

$$\Pi(u) = \frac{1}{2}\int_V \epsilon^T D\epsilon dV - \sum_i u_i^T P_i$$

As the structure in static equilibrium, the total virtual work done by the sum of the internal and external forces is zero. This is represented by the first variation of the total potential energy statement being equal to zero as shown below.

$$\delta\Pi(u) = 0$$

$$\delta\Pi = \delta U_I - \delta U_E = \int_V \epsilon^T D\delta\epsilon dV - \sum_i \delta u_i^T P_i = 0$$

Therefore

$$\int_V \epsilon^T D\delta\epsilon dV = \sum_i \delta u_i^T P_i$$

Where ϵ, u are defined as

$\{u\} = [N]\{d\}$ Where N is the matrix of shape functions

$\{\epsilon\} = [\partial][u]$ Strain is known to be the partial derivatives of displacement

{d} is the elemental degrees of freedom ie $\{u_1 \quad v_1 \quad u_2 \quad v_2\}$

Therefore, substituting u into the epsilon gives

$$\{\epsilon\} = [\partial][N]\{d\}$$

Next, the B matrix is defined as the partial derivatives of the shape functions

$\{B\} = [\partial][N]$ therefore substituting into the strain displacement relationship gives

$$\{\epsilon\} = [B]\{d\}$$

$$\int_V B^T DBdV\,\{d\} = \sum_i \delta u_i^T P_i$$

Therefore, the stiffness matrix is

$$K_e = \int_V B^T DBdV$$

For the 2d case the volume spans x, y and thickness t is assumed to be constant.

$$K_e = t\int_y \int_x [B]^T[D][B]\;dx\;dy$$

3.2.1 B Matrix Derivation

Starting from the strain displacement formula provided previously

$$\{\epsilon\} = [B]\{d\}$$

Strain is the change in displacement with respect to a given direction. Small displacements and small rotations are assumed. Therefore, all higher order terms are ignored.

$$\epsilon_x = \frac{\partial u}{\partial x}, \qquad \epsilon_y = \frac{\partial v}{\partial y}, \qquad \gamma_{xy} = \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)$$

Re-arranging in matrix form

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{Bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial y} \end{Bmatrix}$$

The core logic of the finite element method is to approximate the global (u, v) displacement field with shape functions that interpolate nodal displacements over the range of a single element.

$$\hat{u}(x,y) = \sum_{i=1}^{n} N_i(x,y)u_i\,, \qquad \hat{v}(x,y) = \sum_{i=1}^{n} N_i(x,y)v_i$$

$$\begin{Bmatrix} \hat{u} \\ \hat{v} \end{Bmatrix} = \begin{bmatrix} N_i & 0 \\ 0 & N_i \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$$

Computing the derivatives

$$N_{i,x} = \frac{\partial N_i}{\partial x}$$

$$\begin{Bmatrix} \hat{u}_{,x} \\ \hat{u}_{,y} \\ \hat{v}_{,x} \\ \hat{v}_{,y} \end{Bmatrix} = \begin{bmatrix} N_{i,x} & 0 \\ N_{i,y} & 0 \\ 0 & N_{i,x} \\ 0 & N_{i,y} \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$$

Substituting into the strain displacement relationship

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} N_{i,x} & 0 \\ N_{i,y} & 0 \\ 0 & N_{i,x} \\ 0 & N_{i,y} \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$$

This simplifies to

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} N_{i,x} & 0 \\ 0 & N_{i,y} \\ N_{i,y} & N_{i,x} \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$$

Therefore, the B matrix is as shown. With the index i representing each node within the element.

$$B_i = \begin{bmatrix} N_{i,x} & 0 \\ 0 & N_{i,y} \\ N_{i,y} & N_{i,x} \end{bmatrix}$$

3.3 Boundary Conditions

As all models are assumed to be 2d and remain planar. Only translation constraints within the planar directions (x, y) are necessary.

4 Stiffness Derivation

4.1 Calculation in the Global System

The previously shown elemental stiffness calculation's volume can be calculated across x, y for a quadrilateral element. This approach is feasible for perfectly square elements. However, it becomes challenging to determine the proper integration limits when the element becomes distorted. Virtually every element within a FE model has some level distortion. Thus, the elements will be mapped from the global coordinate systems into an idealized natural coordinate system as shown in Figure 7.

$$[K_e] = t\int_y \int_x [B]^T[D][B]\, dx\, dy$$

4.2 Calculation in the Element Natural Coordinate system

The natural coordinate system for the quadrilateral element is in the ξ, η system with the 4 corners at (-1, -1), (1, -1), (1, 1), (-1, 1). Regardless of the element's physical position, rotation, or skew; it will always map to the same four points.

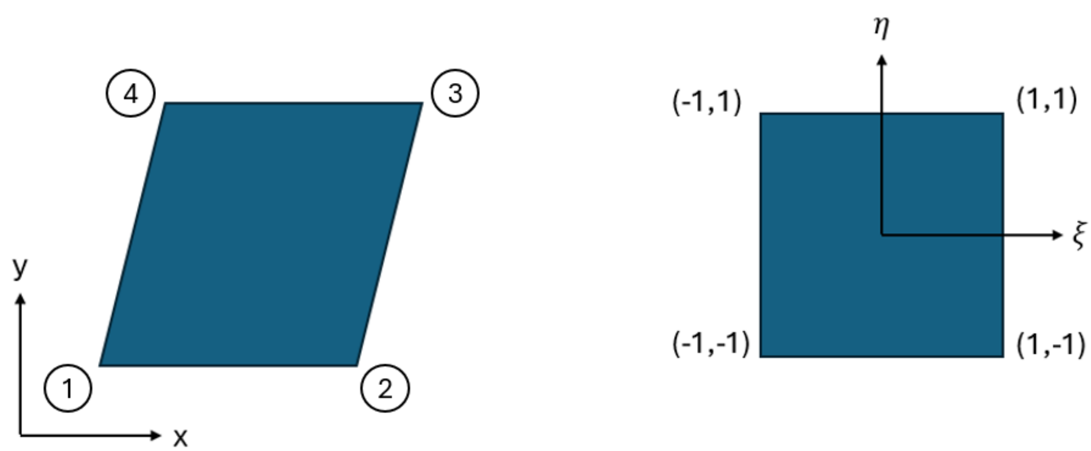


Figure 7: Global to Natural Coordinate System

As shown previously the displacements can be defined as

$$\begin{Bmatrix} u \\ v \end{Bmatrix} = \begin{Bmatrix} \sum N_i(x,y)u_i \\ \sum N_i(x,y)v_i \end{Bmatrix}$$

However, the shape functions here are defined in the x, y coordinate system. The natural coordinate system with the coordinates ξ, η necessitates that the shape functions be functions of ξ, η. An example of these shape functions for a CQUAD4 element are shown below

$N_1 = \frac{1}{4}(1-\xi)(1-\eta),\ N_2 = \frac{1}{4}(1+\xi)(1+\eta)$

$N_3 = \frac{1}{4}(1+\xi)(1+\eta)$, $N_4 = \frac{1}{4}(1-\xi)(1+\eta)$

With the shape functions being in terms of ξ, eta and elemental displacements in terms of x, y; A mapping function from ξ, eta to x, y is required. This is completed with the Jacobian as the Jacobian maps the changes in displacements from natural coordinate system into the global coordinate system with the use of partial derivatives. [3]

$$\begin{bmatrix} x_i \\ y_i \end{bmatrix} = [\,J\,] \begin{bmatrix} \xi_i \\ \eta_i \end{bmatrix}$$

The Jacobian is defined as:

$$[\,J\,] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{bmatrix}$$

And the determinant of the Jacobian is

$$|J| = \frac{\partial x}{\partial \xi}\frac{\partial y}{\partial \eta} - \frac{\partial x}{\partial \eta}\frac{\partial y}{\partial \xi}$$

The bounds of the integration can now be written in-terms of ξ, η with the following relation [3]

$$dxdy = \det[\,J\,]\, d\xi d\eta$$

Therefore

$$[k_e](x, y) = t\int [B]^T[C][B]\det[\,\mathrm{J}]\, d\xi d\eta$$

The B matrix must also be modified to reflect the mapping of the shape functions with the Jacobian.

$$\{\varepsilon\} = [B]\{d\}$$

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{Bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial y} \end{Bmatrix}$$

To get results in terms of ξ, η the multivariable chain rule is used.

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial x}, \qquad \frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial u}{\eta}\frac{\partial \eta}{\partial y}$$

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial v}{\partial \eta}\frac{\partial \eta}{\partial x}, \qquad \frac{\partial v}{\partial y} = \frac{\partial v}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial v}{\eta}\frac{\partial \eta}{\partial y}$$

Re-writing in matrix form

$$\begin{Bmatrix} u_{,x} \\ u_{,y} \\ v_{,x} \\ v_{,y} \end{Bmatrix} = \begin{bmatrix} \xi_{,x} & \eta_{,x} & 0 & 0 \\ \xi_{,y} & \eta_{,y} & 0 & 0 \\ 0 & 0 & \xi_{,x} & \eta_{,x} \\ 0 & 0 & \xi_{,y} & \eta_{,y} \end{bmatrix} \begin{Bmatrix} u_{,\xi} \\ u_{,\eta} \\ v_{,\xi} \\ v_{,\eta} \end{Bmatrix} = \begin{bmatrix} [J]^{-1} & \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} \\ \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} & [J]^{-1} \end{bmatrix} \begin{Bmatrix} u_{,\xi} \\ u_{,\eta} \\ v_{,\xi} \\ v_{,\eta} \end{Bmatrix}$$

Substituting in the strain displacement relation

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \xi_{,x} & \eta_{,x} & 0 & 0 \\ \xi_{,y} & \eta_{,y} & 0 & 0 \\ 0 & 0 & \xi_{,x} & \eta_{,x} \\ 0 & 0 & \xi_{,y} & \eta_{,y} \end{bmatrix} \begin{Bmatrix} u_{,\xi} \\ u_{,\eta} \\ v_{,\xi} \\ v_{,\eta} \end{Bmatrix}$$

$$\hat{u}(\xi,\eta) = \sum_{i=1}^{n} N_i(\xi,\eta) u_i\,, \qquad \hat{v}(\xi,\eta) = \sum_{i=1}^{n} N_i(\xi,\eta) v_i$$

$$\begin{Bmatrix} \hat{u} \\ \hat{v} \end{Bmatrix} = \begin{bmatrix} N_i & 0 \\ 0 & N_i \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$$

Computing the derivatives

$$\begin{Bmatrix} \hat{u}_{,\xi} \\ \hat{u}_{,\eta} \\ \hat{v}_{,\xi} \\ \hat{v}_{,\eta} \end{Bmatrix} = \begin{bmatrix} N_{i,\xi} & 0 \\ N_{i,\eta} & 0 \\ 0 & N_{i,\xi} \\ 0 & N_{i,\eta} \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$$

As the shape functions are functions of ξ, η their derivatives can be taken

Substituting in the final strain displacement relation shows the following where i is the index of nodes

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \xi_{,x} & \eta_{,x} & 0 & 0 \\ \xi_{,y} & \eta_{,y} & 0 & 0 \\ 0 & 0 & \xi_{,x} & \eta_{,x} \\ 0 & 0 & \xi_{,y} & \eta_{,y} \end{bmatrix} \begin{bmatrix} N_{i,\xi} & 0 \\ N_{i,\eta} & 0 \\ 0 & N_{i,\xi} \\ 0 & N_{i,\eta} \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$$

B Matrix:

$$[B] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \xi_{,x} & \eta_{,x} & 0 & 0 \\ \xi_{,y} & \eta_{,y} & 0 & 0 \\ 0 & 0 & \xi_{,x} & \eta_{,x} \\ 0 & 0 & \xi_{,y} & \eta_{,y} \end{bmatrix} \begin{bmatrix} N_{i,\xi} & 0 \\ N_{i,\eta} & 0 \\ 0 & N_{i,\xi} \\ 0 & N_{i,\eta} \end{bmatrix}$$

$$[B] = \begin{bmatrix} \xi_{,x} & \eta_{,x} & 0 & 0 \\ 0 & 0 & \xi_{,y} & \eta_{,y} \\ \xi_{,y} & \eta_{,y} & \xi_{,x} & \eta_{,x} \end{bmatrix} \begin{bmatrix} N_{i,\xi} & 0 \\ N_{i,\eta} & 0 \\ 0 & N_{i,\xi} \\ 0 & N_{i,\eta} \end{bmatrix}$$

$$B = \begin{bmatrix} \xi_{,x} N_{i,\xi} + \eta_{,x} N_{i,\eta} & 0 \\ 0 & \xi_{,y} N_{i,\xi} + \eta_{,y} N_{i,\eta} \\ \xi_{,y} N_{i,\xi} + \eta_{,y} N_{i,\eta} & \xi_{,x} N_{i,\xi} + \eta_{,x} N_{i,\eta} \end{bmatrix}$$

Simplified strain displacement equations

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} \xi_{,x} N_{i,\xi} + \eta_{,x} N_{i,\eta} & 0 \\ 0 & \xi_{,y} N_{i,\xi} + \eta_{,y} N_{i,\eta} \\ \xi_{,y} N_{i,\xi} + \eta_{,y} N_{i,\eta} & \xi_{,x} N_{i,\xi} + \eta_{,x} N_{i,\eta} \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \end{Bmatrix}$$

Returning back to the elemental stiffness calculation, all required variables are now known

$$[k_e](x,y) = t\int\int [B]^T[C][B]\det[\,\mathrm{J}]\, d\xi d\eta$$

4.3 Gauss Quadrature

While programs like python with sympy or Maple can perform symbolic integration, it is inefficient compared to numerical methods. As the integrals consist of the element geometry and shape functions which vary element to element. Thus, Gause integration, a form of numerical integration is used to approximate the integration. Gauss integration says that an integral can be evaluated as a summation of functions evaluated at integration points with weights applied. [4]

$$\int_{-1}^{1} f(x)dx = \sum_{i=1}^{1} w_i f(x_i)$$

For double integration

$$\int_{-1}^{1}\int_{-1}^{1} f(x,y)dxdy = \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j f(x_i, y_i)$$

Substituting into the stiffness calculation

$$[k_e](x,y) = t\sum_{i=1}^{n}\sum_{j=1}^{n}[w_i w_j\, [B]^T[C][B]\det[\,\mathrm{J}]\, d\xi d\eta]$$

The Gauss Points and Weights shown in Figure 8 are selected by the polynomial order of the element's shape functions. Shown in Table 1.

Number of points, *n*	**Points, x_i**	**Weights, w_i**
1	0	2
2	$\pm\sqrt{\frac{1}{3}}$	1
3	0	$\frac{8}{9}$
	$\pm\sqrt{\frac{3}{5}}$	$\frac{5}{9}$

Figure 8: Gauss Quadrature Weights and Points [4]

Table 1: Gauss Points and Weights Per Element Type

Element	Shape Function (Ex)	Gauss Points	Gauss Weights
CTRIA3	$1-\xi-\eta$	$1/3$	0.5
CQUAD4	$0.25*(1-\xi)(1-\eta)$	$\pm\frac{\sqrt{1}}{3}$	1
CQUAD6	$\frac{1}{4}[(1-\xi)(1-\eta)-\ (1-\xi^2)(1-\eta)]$	$\pm\frac{\sqrt{1}}{3}$	1
CQUAD7	$0.25*\xi(1-\xi)(\eta-1)$	$-\sqrt{\frac{3}{5}},0,\sqrt{\frac{3}{5}}$	$\frac{5}{9},\frac{8}{9},\frac{5}{9}$
CQUAD8	$-\frac{1}{4}(1-\eta)(1-\xi)(1+\xi+\eta)$	$-\sqrt{\frac{3}{5}},0,\sqrt{\frac{3}{5}}$	$\frac{5}{9},\frac{8}{9},\frac{5}{9}$

5 Displacement Calculation

The global stiffness matrix is assembled by combining the individual stiffness matrices of each element. There are two degrees of freedom per node, therefore the global stiffness matrix will be of size (2n x 2n)

$$[K]\{u\} = \{p\}$$

The boundary conditions are applied with each fixe dof, removing the corresponding row and columns from the stiffness matrix and applied load vector. With the reduced stiffness matrix the global nodal displacements can by calculated by taking the inverse of the reduced stiffness matrix multiplied by the load vector.

$$\{u\} = [K]^{-1}\{p\}$$

6 Strain and Stress Calculation

With all the global displacements calculated previously, the strain displacement relations of Section 4.2 can be used to calculate the strain and stress at each node for each element. For example for a CQUAD4 the strain calculation would be as follows.

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \xi_{,x} & \eta_{,x} & 0 & 0 \\ \xi_{,y} & \eta_{,y} & 0 & 0 \\ 0 & 0 & \xi_{,x} & \eta_{,x} \\ 0 & 0 & \xi_{,y} & \eta_{,y} \end{bmatrix} \begin{bmatrix} N_{1,\xi} & 0 & N_{2,\xi} & 0 & N_{3,\xi} & 0 & N_{4,\xi} & 0 \\ N_{1,\eta} & 0 & N_{2,\eta} & 0 & N_{3,\eta} & 0 & N_{4,\eta} & 0 \\ 0 & N_{1,\xi} & 0 & N_{2,\xi} & 0 & N_{3,\xi} & 0 & N_{4,\xi} \\ 0 & N_{1,\eta} & 0 & N_{2,\eta} & 0 & N_{3,\eta} & 0 & N_{4,\eta} \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{Bmatrix}$$

As shown previously the shape functions are defined in the natural coordinate system (ξ, η). Therefore, strain is evaluated at each (ξ, eta) point that corresponds to a nodal location of the element. As each element has different mapping between the global and natural coordinate systems, nodal stress results will be unique to that element. Therefore, nodes that are shared by multiple elements will have multiple differing strain values. For plotting, the average of the node's strain values from all adjoining elements is used.

From the constitutive equations stress can be determined with the known nodal strains. With the plate thickness of 0.25" it is assumed that the plate is acting in plane stress. Therefore, the following relationship is used.

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1 - \nu}{2} \end{bmatrix} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{Bmatrix}$$

7 Element Shape Functions

Shape functions, regardless of element type, are used to approximate the global displacement field over the individual element by interpolating the nodal displacements. All shape functions must satisfy two conditions: Kronecker delta, and the partition of unity. The Kronecker Delta states that the shape function must equal one at it's respective node and 0 at all other nodes. The partition of unity states that all points over the space the sum of the shape functions must equal one. The derivation of all shape functions and their derivatives is performed symbolically using the sympy library of python. See Section 12 for more information on the supporting documentation.

7.1 CTRIA3

The CTRIA3 element is the classic Constant Strain Triangle (CST) element seen in literature [3]. As shown in Table 2, the shape functions are linear and therefore have constant derivatives. As such the B matrix shown in Section 4.2 contains only the constant derivatives, resulting in constant strain throughout the element.

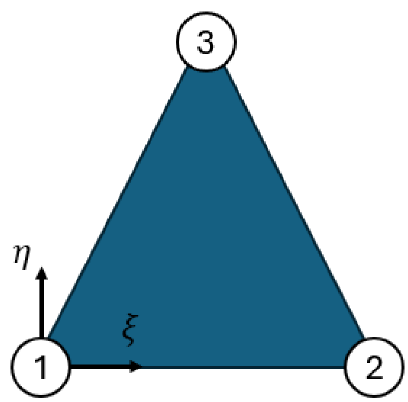


Figure 9: CTRIA3 Element

Table 2: CTRIA3 Shape Functions

Nodal Point	Shape Function	$\frac{\partial N}{\partial \xi}$	$\frac{\partial N}{\partial \eta}$
N1: (0,0)	$1 - \xi - \eta$	-1	-1
N2: (1, 0)	ξ	1	0
N3: (0, 1)	η	0	1

7.2 CQUAD4

The CQUAD 4 element has four corner nodes with CCW ordering starting from the lower left corner as shown in Figure 10. The shape functions for the CQUAD4 element shown in Table 3 are bilinear. The derivatives of the bilinear shape functions result in linear functions. Thus, the strain varies linearly within the element. This can lead to an un-representative stress distribution if the represented area has a nonlinear stress distribution.

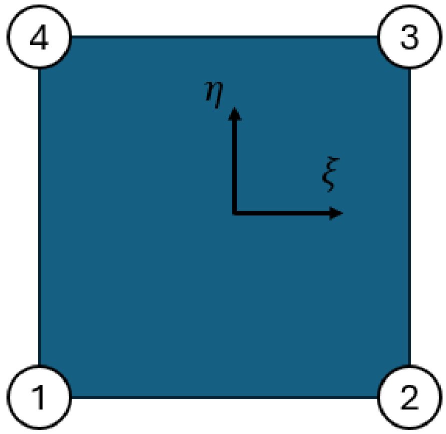


Figure 10: CQUAD4 Element

Table 3: CQUAD4 Shape Functions

Nodal Point	Shape Function	$\frac{\partial N}{\partial \xi}$	$\frac{\partial N}{\partial \eta}$
N1: (-1, -1)	$\frac{1}{4}(1-\xi)(1-\eta)$	$0.25\eta - 0.25$	$0.25\xi - 0.25$
N2: (1, -1)	$\frac{1}{4}(1+\xi)(1-\eta)$	$0.25 - 0.25\eta$	$-0.25\xi - 0.25$
N3: (1, 1)	$\frac{1}{4}(1+\xi)(1+\eta)$	$0.25\eta + 0.25$	$0.25\xi + 0.25$
N4: (-1, 1)	$\frac{1}{4}(1-\xi)(1+\eta)$	$-0.25\eta - 0.25$	$0.25 - 0.25\xi$

7.3 CQUAD8

The CQUAD8 element has four corner nodes with CCW ordering starting from the lower left corner with the second set of four nodes found at the corresponding midpoints between the corner nodes as shown in Figure 11. As shown in Table 4 the shape functions for all points use quadratic interpolation.

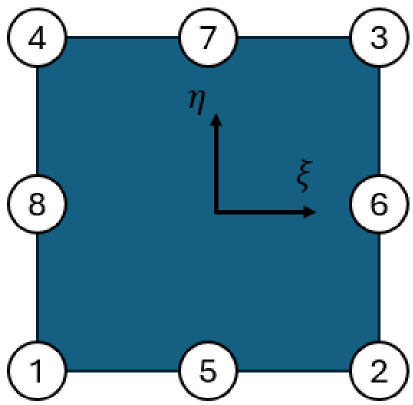


Figure 11: CQUAD8 Element

Table 4: CQUAD8 Shape Functions

Nodal Point	Shape Function	$\frac{\partial N}{\partial \xi}$	$\frac{\partial N}{\partial \eta}$
N1: (-1, -1)	$-\frac{1}{4}(1-\eta)(1-\xi)(1+\xi+\eta)$	$0.25(-\eta-2\xi)(\eta-1)$	$0.25(-2\eta-\xi)(\xi-1)$
N2: (1, -1)	$-\frac{1}{4}(1-\eta)(1+\xi)(1-\xi+\eta)$	$0.25(\eta-1)(\eta-2\xi)$	$0.25(2\eta-\xi)(\xi+1)$
N3: (1, 1)	$-\frac{1}{4}(1+\eta)(1+\xi)(1-\xi-\eta)$	$0.25(\eta+1)(\eta+2\xi)$	$0.25(2\eta+\xi)(\xi+1)$
N4: (-1, 1)	$-\frac{1}{4}(1+\eta)(1-\xi)(1+\xi-\eta)$	$0.25(-\eta+2\xi)(\eta+1)$	$0.25(-2\eta+\xi)(\xi-1)$
N5: (0, -1)	$\frac{1}{2}(1-\eta)(1-\xi)(1+\xi)$	$\xi(\eta-1)$	$0.5\xi^2-0.5$
N6: (1, 0)	$\frac{1}{2}(1-\eta)(1+\xi)(1+\eta)$	$0.5-0.5\eta^2$	$\eta(-\xi-1)$
N7: (0, 1)	$\frac{1}{2}(1+\eta)(1-\xi)(1+\xi)$	$\xi(-\eta-1)$	$0.5-0.5\xi^2$
N8: (-1, 0)	$\frac{1}{2}(1-\eta)(1-\xi)(1+\eta)$	$0.5\eta^2-0.5$	$\eta(\xi-1)$

7.4 CQUAD7

The CQUAD7 while similar to the CQUAD8 shown in Figure 11 has a missing node in the 8th position. Thus, the shape functions (Table 5) have no obligation to be 1 or 0 at that point to satisfy the Kronecker delta property. This element allows for three CQUAD8 elements to be attached and transition to a CQUAD4 element. For this transition to occur the stress distribution at this edge must match that of the CQUAD4 element. Thus, the edge between nodes 1 & 4 is linearly interpolated like that of the CQUAD4 while the other edges retain the quadratic interpolation similar to the CQUAD8.

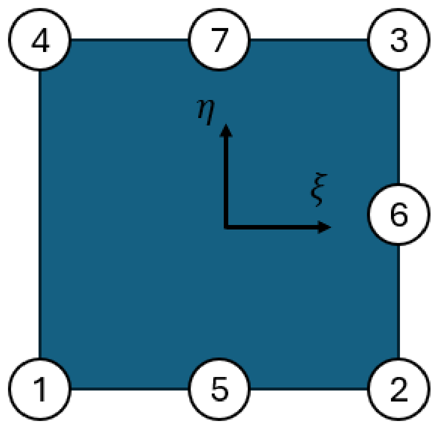


Figure 12: CQUAD7 Element

Table 5: CQUAD7 Shape Functions

Nodal Point	Shape Function	$\frac{\partial N}{\partial \xi}$	$\frac{\partial N}{\partial \eta}$
N1: (-1, -1)	$\frac{1}{4}(1-\xi)(1-\eta) - \frac{1}{2}N_5$	$0.25\eta - 0.5\xi(\eta-1) - 0.25$	$0.25\xi(1-\xi)$
N2: (1, -1)	$\frac{1}{4}(1+\xi)(1-\eta) - \frac{1}{2}N_5 - \frac{1}{2}N_6$	$0.25\eta^2 - 0.25\eta - 0.5\xi(\eta-1)$	$0.5\eta(\xi+1) - 0.25\xi^2 - 0.25\xi$
N3: (1, 1)	$\frac{1}{4}(1+\xi)(1+\eta) - \frac{1}{2}N_6 - \frac{1}{2}N_7$	$0.25\eta^2 + 0.25\eta + 0.5\xi(\eta+1)$	$0.5\eta(\xi+1) + 0.25\xi^2 + 0.25\xi$
N4: (-1, 1)	$\frac{1}{4}(1-\xi)(1+\eta) - \frac{1}{2}N_7$	$-0.25\eta + 0.5\xi(\eta+1) - 0.25$	$0.25\xi(\xi-1)$
N5: (0, -1)	$\frac{1}{2}(1-\xi^2)(1-\eta)$	$\xi(\eta-1)$	$0.5\xi^2 - 0.5$
N6: (1, 0)	$\frac{1}{2}(1+\xi)(1-\eta^2)$	$0.5 - 0.5\eta^2$	$-\eta(\xi+1)$
N7: (0, 1)	$\frac{1}{2}(1-\xi^2)(1+\eta)$	$-\xi(\eta+1)$	$0.5 - 0.5\xi^2$

Evaluating the shape functions shown in Table 5 along the edge at $(-1, \eta)$ shows that only N1 and N4 produce non-zero equations with the functions being linear as required by the adjoining CQUAD4 element.

$$N1 = 0.5 - 0.5\eta, \qquad N2 = 0.5\eta + 0.5$$

7.5 CQUAD6

Similar to the CQUAD 7, the CQUAD6 has two missing midside nodes as shown in Figure 13. This allows for two CQUAD8 elements to transition to two CQUAD4 using the shape functions shown in Table 6.

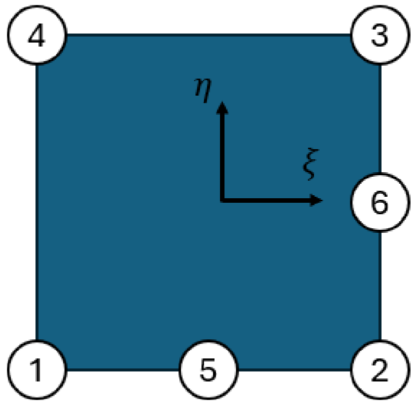


Figure 13: CQUAD6 Element

Table 6: CQUAD6 Shape Functions

Nodal Point	Shape Function	$\frac{\partial N}{\partial \xi}$	$\frac{\partial N}{\partial \eta}$
N1: (-1, -1)	$\frac{1}{4}(1-\xi)(1-\eta)-\frac{1}{2}N_5$	$0.25\eta-0.5\xi(\eta-1)-0.25$	$0.25\xi(1-\xi)$
N2: (1, -1)	$\frac{1}{4}(1+\xi)(1-\eta)-\frac{1}{2}N_5-\frac{1}{2}N_6$	$0.25\eta^2-0.25\eta-0.5\xi(\eta-1)$	$0.5\eta(\xi+1)-0.25\xi^2-0.25\xi$
N3: (1, 1)	$\frac{1}{4}(1+\xi)(1+\eta)-\frac{1}{2}N_6$	$0.25\eta(\eta+1)$	$0.5\eta(\xi+1)+0.25\xi^2+0.25\xi$
N4: (-1, 1)	$\frac{1}{4}(1-\xi)(1+\eta)$	$-0.25\eta-0.25$	$0.25\xi(\xi-1)$
N5: (0, -1)	$\frac{1}{2}(1-\xi^2)(1-\eta)$	$\xi(\eta-1)$	$0.5\xi^2-0.5$
N6: (1, 0)	$\frac{1}{2}(1+\xi)(1-\eta^2)$	$0.5-0.5\eta^2$	$-\eta(\xi+1)$

The linear relationship between nodes 1 and 4 remain unchanged while the relationship between nodes 3 and 4 becomes linear to accommodate the second CQUAD 4 element. This is shown when the shape functions are evaluated at $(\xi, 1)$

$$N3 = 0.5\xi + 0.5, \qquad N4 = 0.5 - 0.5\xi$$

8 Python Solver Validation

To validate the Python finite element solver, a model of a large plate with an unloaded hole is used. This plate has tensile loading applied at the right edge. The plate is constrained such that all nodes except the center line node along the left edge are free to displace in the y direction. Thus, the structure will not resist the poisons effect due to the applied tension load. The center line node along the left edge is fully fixed to eliminate rigid body motion of the full model. This model can be see in Figure 14.

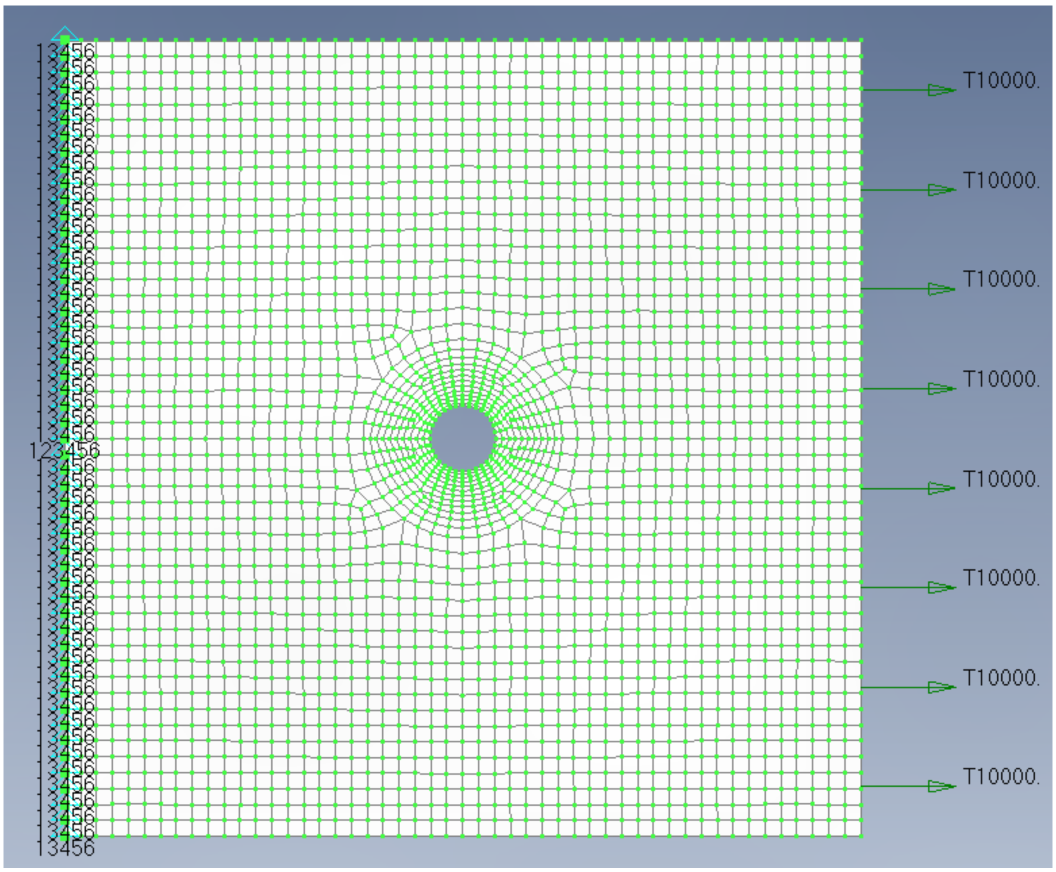


Figure 14: Tension Plate

8.1 Elasticity Solution

With the modeled plate being large relative to the hole diameter, the plate can be assumed to be infinite. Thus, the classical Krisch solution for an infinite plate with a circular hole under tension can be used. The polar stress field is defined below. [5]

$$\sigma_{rr} = \frac{\sigma}{2}\left(1 - \frac{a^2}{r^2}\right) + \frac{\sigma}{2}\left(1 + 3\frac{a^4}{r^4} - 4\frac{a^2}{r^2}\right)cos2\theta$$

$$\sigma_{\theta\theta} = \frac{\sigma}{2}\left(1 + \frac{a^2}{r^2}\right) - \frac{\sigma}{2}\left(1 + 3\frac{a^4}{r^4}\right)cos2\theta$$

$$\sigma_{r\theta} = -\frac{\sigma}{2}\left(1 - 3\frac{a^4}{r^4} + 2\frac{a^2}{r^2}\right)\sin 2\theta$$

The critical stress occurs when r = a and $\theta = \frac{\pi}{2}$

$$\sigma_{rr} = \frac{\sigma}{2}\left(1 - \frac{1}{1}\right) + \frac{\sigma}{2}(1 + 3 - 4)cos\pi = 0$$

$$\sigma_{\theta\theta} = \frac{\sigma}{2}(1 + 1) - \frac{\sigma}{2}(1 + 3)cos\pi = \sigma - (-2\sigma) = 3\sigma$$

$$\sigma_{r\theta} = -\frac{\sigma}{2}(1 - 3 + 2)\sin\pi = 0$$

This closed form solution provides the exact solution to this problem and shows the familiar Kt of 3 which occurs at $\theta \pm \pi/2$. This is an exact solution of which FEM solvers can be compared against to determine validity.

8.2 FEMAP Solution

Creating this model within FEMAP with a highly refined CQ4 mesh produces the expected max principal stress distribution as shown in Figure 15. This model has 10000 lbf load applied and an area of 1.5in^2. Calculating the stresses below shows that FEMAP is matching the exact elasticity solution as is expected of a commercially available FEM suite.

$$\sigma_{gross} = \frac{1000\ lbf}{6in * 0.25in} = 6.67\ ksi$$

$$\sigma_{max} = 20.086\ ksi$$

$$Kt = \frac{\sigma_{max}}{\sigma_{gross}} = \frac{20.086}{6.67} = 3.01$$

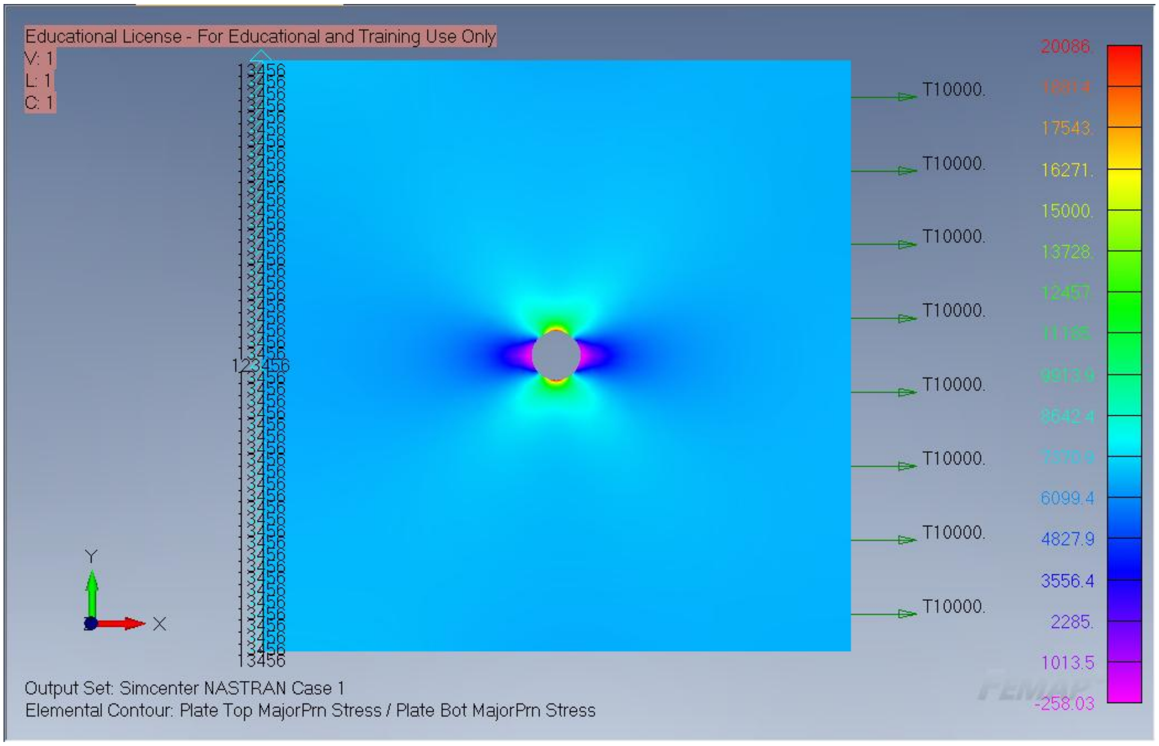


Figure 15: Femap Solution for Open Hole Tension Plate

8.3 Python Solver

Figure 16 shows the Max Principal Stress fringe plot for the same mesh as used by FEMAP containing only CQ4 elements. The maximum principal stress is 20.373 ksi, 1.9% difference from the elasticity solution. This difference is within reasonable limits to consider the PySolver as validated for CQ4 elements. Figure 17 shows the same model with a course mesh of CQ4 elements and a refined region of CQ8 elements around the hole area of interest with CQ7 elements spanning the transition boundary. The results of this model shown in Figure 18 produced a maximum principal stress of 19.347 ksi. Considering the coarseness of this mesh an error 3.26% is reasonable to consider this solver validated for CQ8 and CQ7 elements. Refining the mesh further in the transition region improves correlation with a stress of 19.624 ksi and an error of 1.8%

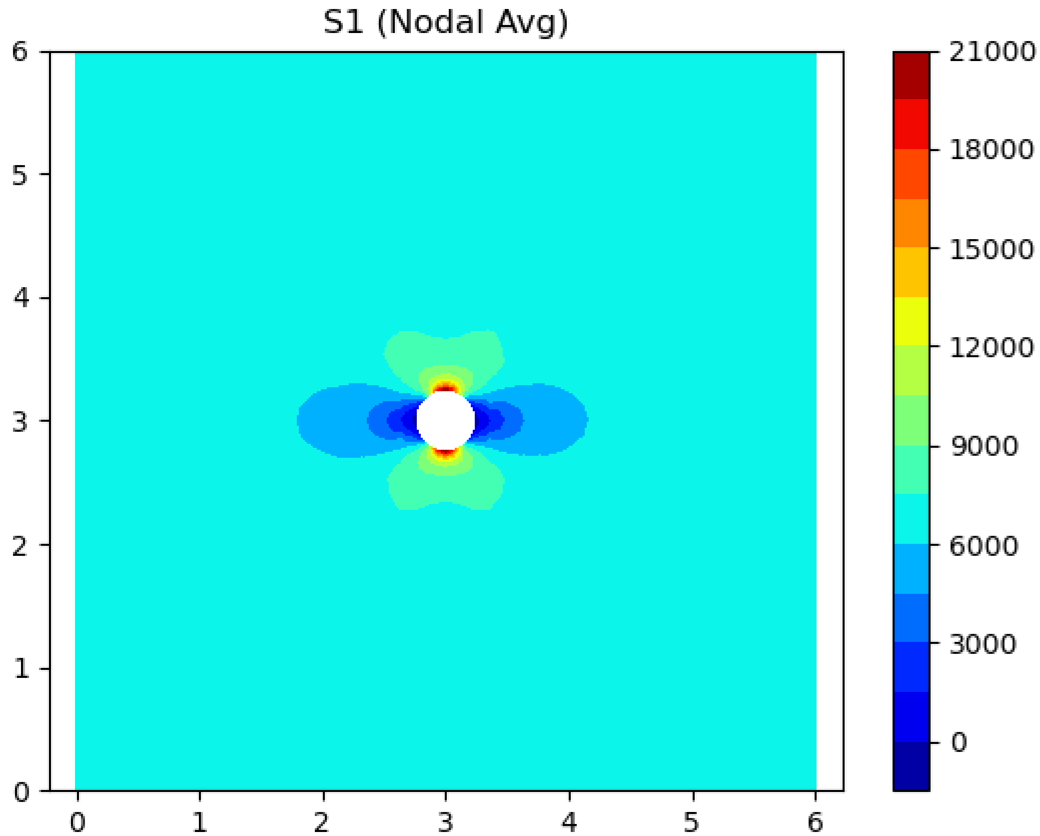


Figure 16: Python Solver for Open Hole Plate

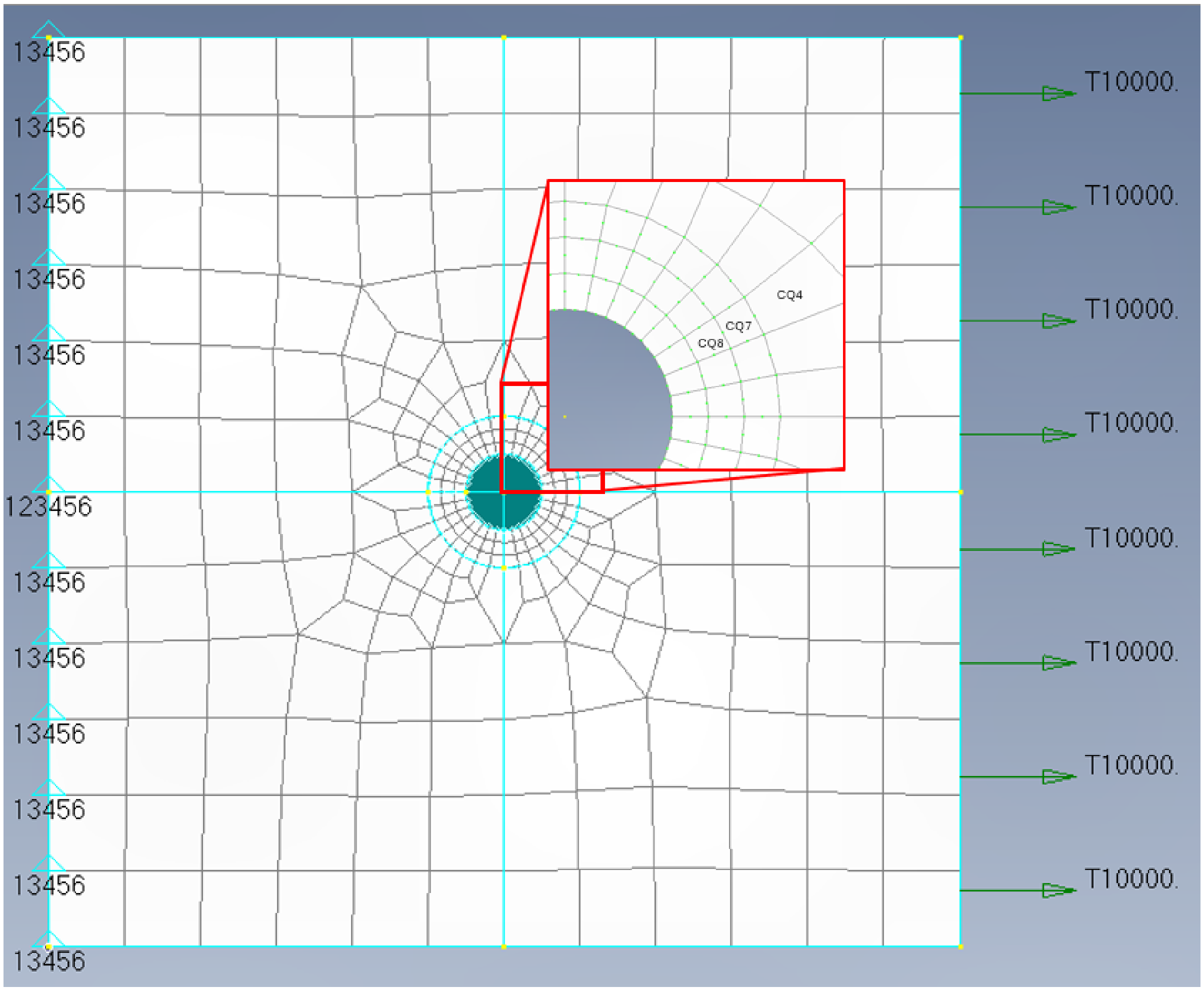


Figure 17: CQ8/CQ7 Open Hole Plate Model

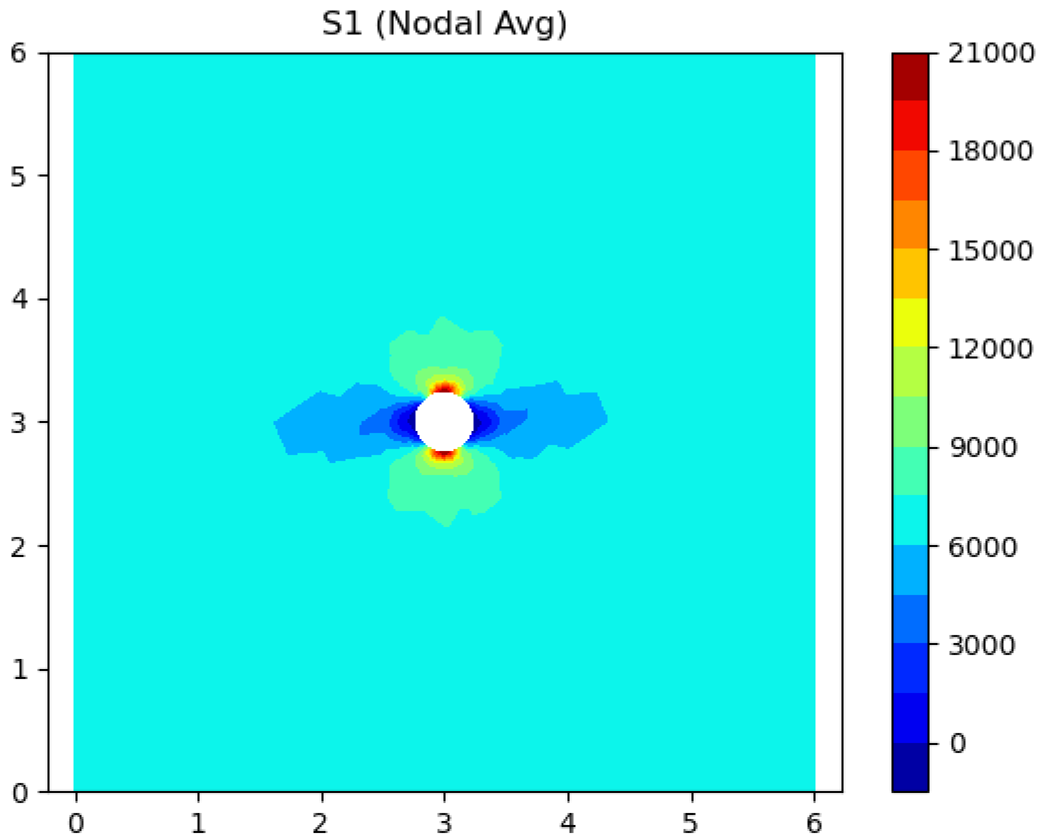


Figure 18: CQ8/CQ7 Open Hole Plate Results

9 Stress Concentration Determination

From the initial problem statement, the initial mesh of the critical region shown in Figure 4 is not capable of capturing the Kt nor does the geometry shown in Figure 5 match that of the classical shoulder figure Kt shown in Figure 6 [2]. To determine the Kt value due to shoulder fillet and cap runout a refined mesh of both CQ4s, and CQ8/7/6s was created. Refinement continued until stress convergence was reached for one method. The transition CQ7/CQ6 elements are done outside of the critical area to minimize the influence of the transition between quadratic and bilinear elements.

The plate was subjected to a 1000 lbf of tension load at the right edge. The lower left corner node is fixed in both x and y degrees of freedom to prevent rigid body motion. The left edge has x displacement restrained with free y displacement. It is assumed that the surrounding structure of the vertical frame is sufficiently stiff to eliminate displacement in the y direction. Thus, the y degree of freedom for the lower edge is constrained.

All models use the same 2024 Aluminum with a thickness of 0.25”. As shown in Figure 5 the width of the reduced section is 0.728”. Thus, the gross stress is:

$$\sigma_{Gross} = \frac{1000\ lbf}{0.25\ in * 0.728\ \text{in}} = \ 5495\ psi$$

9.1 Stress Convergence

Figure 19 shows the general plate model with the region highlighted in red being the refinement region. All model names will follow the pattern of CQX_Y. Where X is either 4 or 8 signifying if the refinement region consists of CQ4 or CQ8 elements. Y signifies how many elements are within the fillet radius. Figure 20 shows the transition region of the mesh. The free-disconnected nodes between the CQ8’s and the CQ4’s are automatically detected and removed, converting them to a CQ7 or CQ6 element. Figure 21, and Figure 22 shows how the critical region was refined with increasing mesh density.

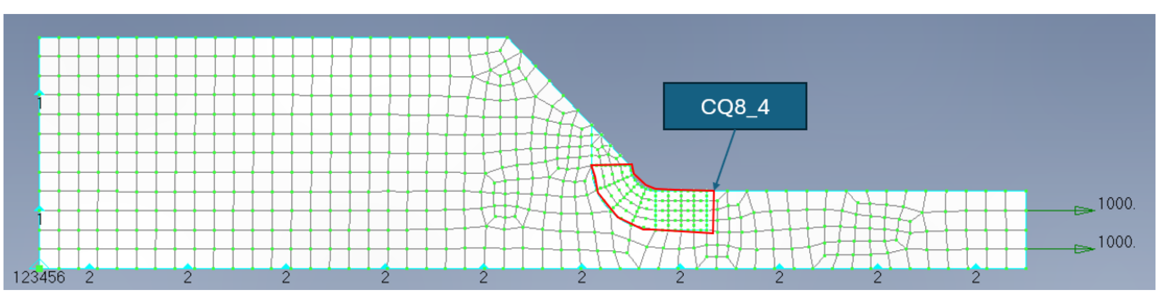


Figure 19: CQ8_4 Model

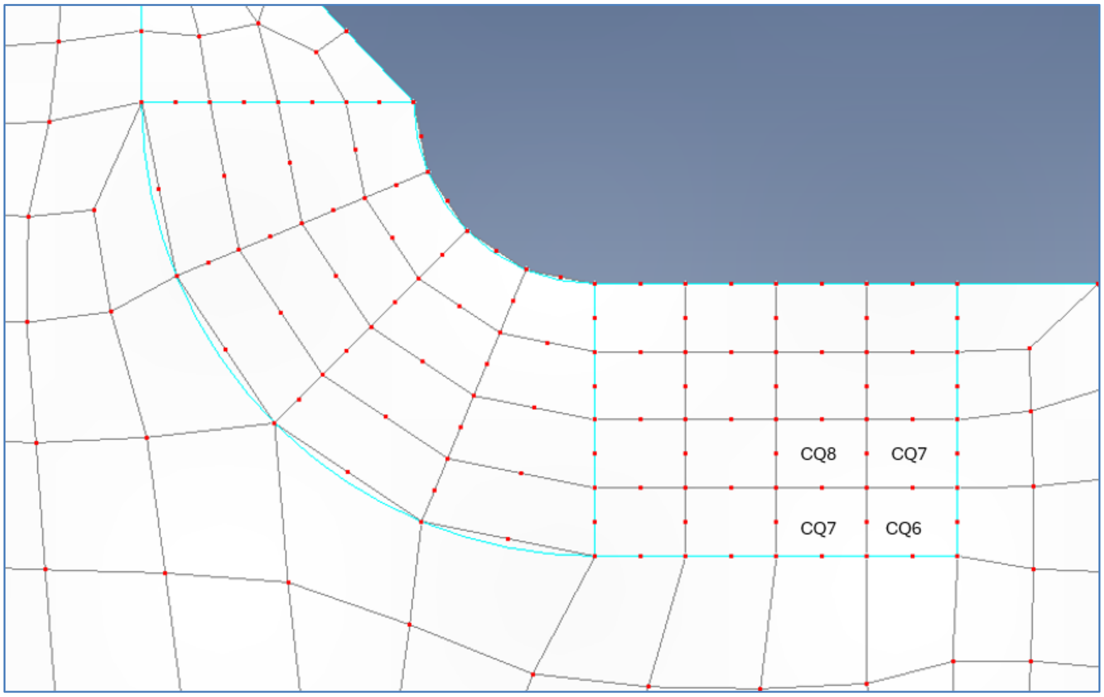


Figure 20: CQ8_4 Transition Region

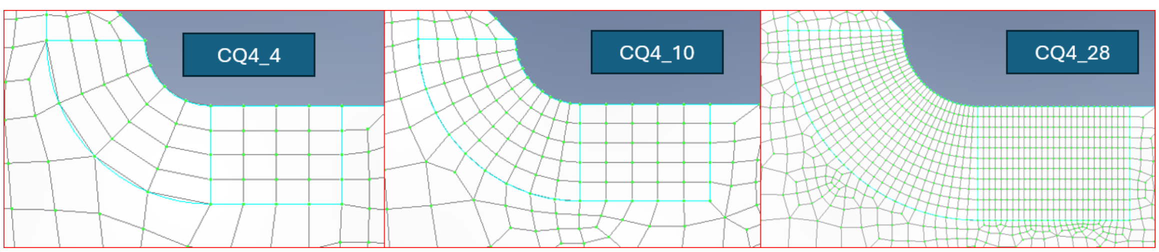


Figure 21: CQ4 Refinement Region

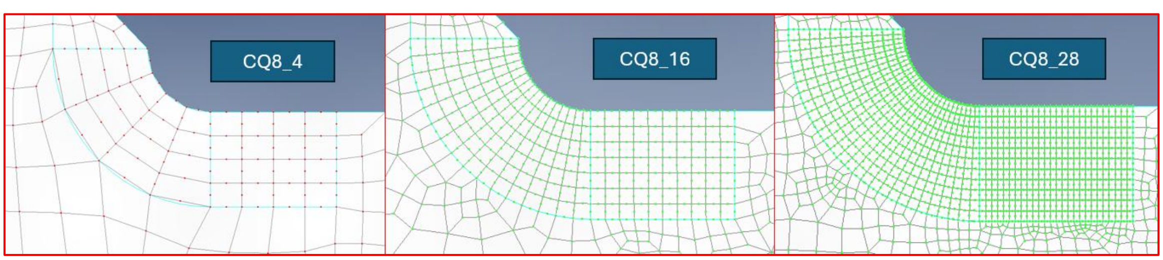


Figure 22: CQ8 Region Refinement Region

Table 7: CQ4 Stress Convergence

Model	Element	NID	S1	S2	S12	S_max	S_min	% Difference
CQ4_4	4	5	10401	1523	-1404	10456	1468	
CQ4_7	493	547	10928	2536	-2450	11103	2361	5.8%
CQ4_10	969	960	11007	2087	-2089	11128	1966	0.2%
CQ4_13	1544	1468	11000	2471	-2586	11191	2280	0.6%
CQ4_16	2161	2021	11111	1978	-2304	11254	1835	0.6%
CQ4_18	3740	3429	11060	2073	-2780	11270	1863	0.1%
CQ4_22	11701	9515	11148	1684	-2414	11300	1533	0.3%
CQ4_28	13941	13338	11124	1728	-2495	11287	1565	-0.1%

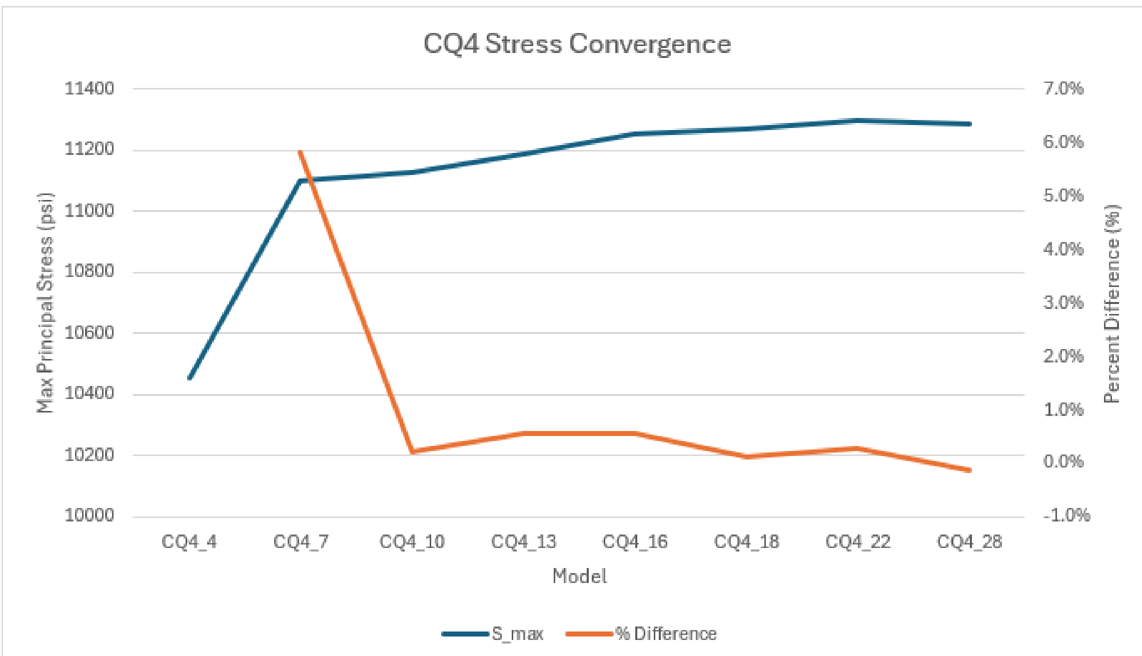


Figure 23: CQ4 Stress Convergence

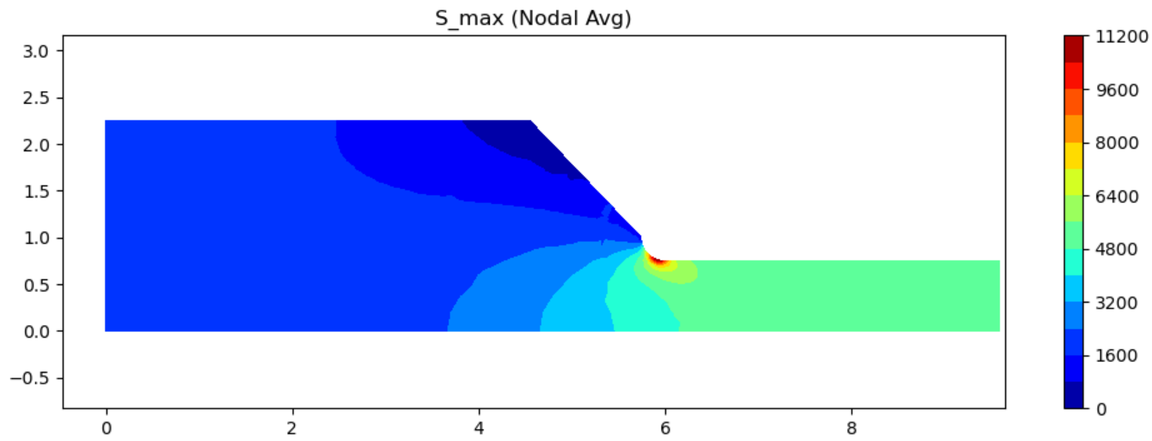


Figure 24: CW4_16 Max Principal Stress(psi)

Table 8: CQ8 Stress Convergence

Model	Element	NID	S1	S2	S12	S_max	S_min	% Difference
CQ8_4	999	6	10504	2192	-3880	10935	1762	
CQ8_7	1481	547	11065	1291	-2593	11234	1122	2.7%
CQ8_10	2146	961	10773	1593	-3306	11061	1304	-1.6%
CQ8_13	2754	1468	10966	1160	-2628	11139	987	0.7%
CQ8_16	3718	2021	10963	760	-2165	11076	646	-0.6%
CQ8_18	4875	3429	10866	1001	-2856	11068	798	-0.1%
CQ8_22	9851	9515	10939	689	-2358	11073	555	0.0%
CQ8_28	11701	13338	10919	735	-2469	11067	587	-0.1%

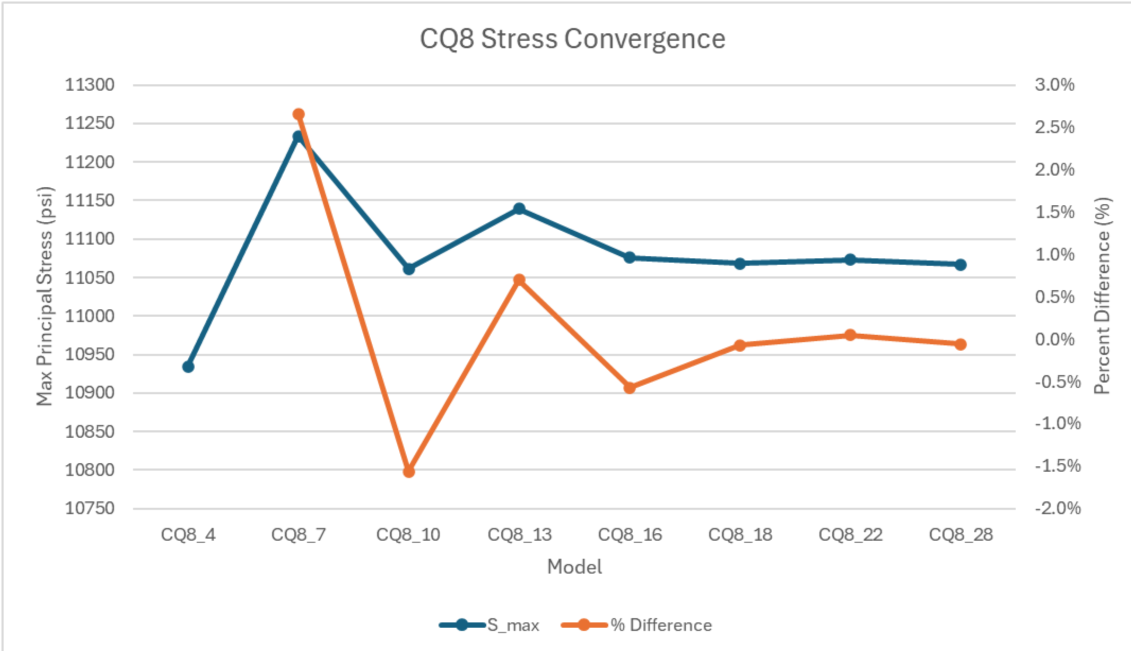


Figure 25: CQ8 Stress Convergence

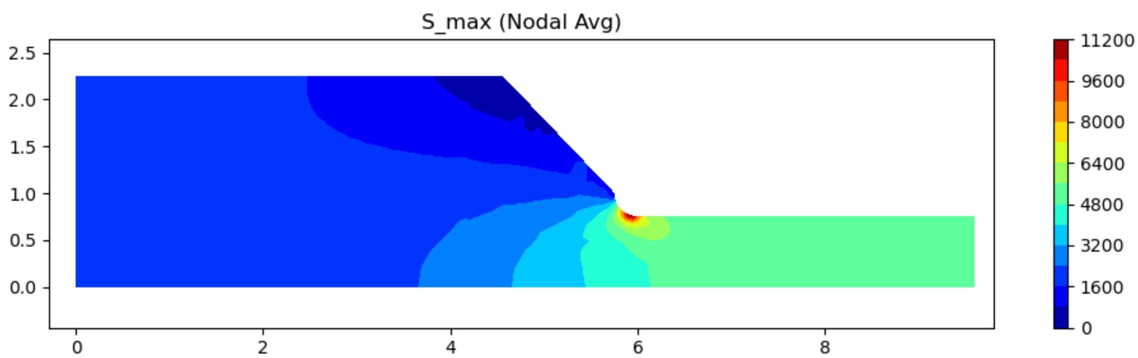


Figure 26: CQ8_10 Max Principal Stress (psi)

9.2 Final Stress Concentration

Figure 23 shows that the CQ4 refined model converges with the CQ4_18 model with a stress level of 11.270 ksi shown in Table 7. Conversely, Figure 23 and Table 8 show that the CQ8 model converged sooner with the CQ8_10 model with a stress level of 11.061 ksi. As such the converged stress of 11.061 ksi will be used to determine the final Kt value.

$$Kt = \frac{\sigma_{max}}{\sigma_{gross}}$$

$$\sigma_{gross} = \frac{P}{A} = \frac{1000}{(0.75 * 0.25)} = 5333\ psi$$

$$Kt = \frac{11.061}{5.333} = 2.07$$

The final converged Kt of 2.07 is 14% greater than the un-conservative assumption and 80% of the over-conservative option. With this final Kt computed an accurate LEFM analysis model could be created to analyze this critical through thickness edge crack detail. An additional benefit of the Python solver over a program like FEMAP was the ability to export nodal stress values. This allows for stresses at the free edge to be pulled rather than centroidal element value.

10 Conclusions

In conclusion, the Python based FEM solver was demonstrated to be able to accurately calculate displacements, strains, and stresses for models containing standard CTRIA3, CQUAD4, CQUAD8, and nonstandard CQUAD6/7 elements as shown in Section 8. The CQUAD8 models with transition elements does converge sooner than the CQUAD4 models.

However, for this analysis the increased convergence speed only saves 98 degrees of freedom at the cost of an additional 960 calculations due to the greater number of integration points from 4 to 9 points with the CQ8 elements.

While the CQUAD8 model with transition elements is a valid and feasible approach, the increased complexity, and calculations provide little benefit over the CQUAD4 approach. Thus, it is generally recommended to use the simpler, standard CQUAD4 approach.

11 References

[1] B. L. S. a. W. J. H. Horn, Failure, Fracture Mechanics, Fatigue, and Damage Tolerance, Wichita State University Department of Aerospace Engineering.

[2] W. D. Pilkey, Perterson's Stress Concentration Factors, John Wiley & Sons, Inc.

[3] R. D. Cook, Concepts and Applications of Finite Element Analysis, John Wiley & Sons, 1989.

[4] "UPC.edu," [Online]. Available: https://numfactory.upc.edu/web/Calculo2/P2_Integracio/html/Gauss1D.html.

[5] Timoshenko, Theory of Elasticity, McGraw-Hill, 1970.

12 Appendix 1: Electronic Files

```
C:.
│   CEM_Solver_MAIN.py
│
├───!Report
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│       Report_Figures.pptx
│
├───.vscode
│       settings.json
│
├───Lib
│   │   DatFileParser.py
│   │   DispCalc.py
│   │   ElementRepository.py
│   │   FringePlotter.py
│   │   KeCalc.py
│   │   StressStrainCalc.py
│   │   __init__.py
│   │
│   └───__pycache__
│           DatFileParser.cpython-312.pyc
│           DispCalc.cpython-312.pyc
│           ElementRepository.cpython-312.pyc
│           FringePlotter.cpython-312.pyc
│           KeCalc.cpython-312.pyc
│           StressStrainCalc.cpython-312.pyc
│
```

```
├───Nastran
│   ├───!FinalModels
│   │   ├───CompoundTensionStrip3
│   │   │   ├───CQ4
│   │   │   │       CompoundTensionStrip_CQ4_10-0000.dat
│   │   │   │       compoundtensionstrip_cq4_10-0000.f04
│   │   │   │       compoundtensionstrip_cq4_10-0000.f06
│   │   │   │       compoundtensionstrip_cq4_10-0000.log
│   │   │   │       compoundtensionstrip_cq4_10-0000.mon1
│   │   │   │       compoundtensionstrip_cq4_10-0000.xdb
│   │   │   │       CompoundTensionStrip_CQ4_10.modfem
│   │   │   │       CompoundTensionStrip_CQ4_13-0000.dat
│   │   │   │       compoundtensionstrip_cq4_13-0000.f04
│   │   │   │       compoundtensionstrip_cq4_13-0000.f06
│   │   │   │       compoundtensionstrip_cq4_13-0000.log
│   │   │   │       compoundtensionstrip_cq4_13-0000.mon1
│   │   │   │       compoundtensionstrip_cq4_13-0000.xdb
│   │   │   │       CompoundTensionStrip_CQ4_13.modfem
│   │   │   │       CompoundTensionStrip_CQ4_16-0000.dat
│   │   │   │       compoundtensionstrip_cq4_16-0000.f04
│   │   │   │       compoundtensionstrip_cq4_16-0000.f06
│   │   │   │       compoundtensionstrip_cq4_16-0000.log
│   │   │   │       compoundtensionstrip_cq4_16-0000.mon1
│   │   │   │       compoundtensionstrip_cq4_16-0000.xdb
│   │   │   │       CompoundTensionStrip_CQ4_16.modfem
│   │   │   │       CompoundTensionStrip_CQ4_18-0000.dat
│   │   │   │       compoundtensionstrip_cq4_18-0000.f04
│   │   │   │       compoundtensionstrip_cq4_18-0000.f06
```

```
|   |   |   |   compoundtensionstrip_cq4_18-0000.log
|   |   |   |   compoundtensionstrip_cq4_18-0000.mon1
|   |   |   |   compoundtensionstrip_cq4_18-0000.xdb
|   |   |   |   CompoundTensionStrip_CQ4_18.modfem
|   |   |   |   CompoundTensionStrip_CQ4_22-0000.dat
|   |   |   |   compoundtensionstrip_cq4_22-0000.f04
|   |   |   |   compoundtensionstrip_cq4_22-0000.f06
|   |   |   |   compoundtensionstrip_cq4_22-0000.log
|   |   |   |   compoundtensionstrip_cq4_22-0000.mon1
|   |   |   |   compoundtensionstrip_cq4_22-0000.xdb
|   |   |   |   CompoundTensionStrip_CQ4_22.modfem
|   |   |   |   CompoundTensionStrip_CQ4_28-0000.dat
|   |   |   |   compoundtensionstrip_cq4_28-0000.f04
|   |   |   |   compoundtensionstrip_cq4_28-0000.f06
|   |   |   |   compoundtensionstrip_cq4_28-0000.log
|   |   |   |   compoundtensionstrip_cq4_28-0000.mon1
|   |   |   |   compoundtensionstrip_cq4_28-0000.xdb
|   |   |   |   CompoundTensionStrip_CQ4_28.modfem
|   |   |   |   CompoundTensionStrip_CQ4_4-0000.dat
|   |   |   |   compoundtensionstrip_cq4_4-0000.f04
|   |   |   |   compoundtensionstrip_cq4_4-0000.f06
|   |   |   |   compoundtensionstrip_cq4_4-0000.log
|   |   |   |   compoundtensionstrip_cq4_4-0000.mon1
|   |   |   |   compoundtensionstrip_cq4_4-0000.xdb
|   |   |   |   CompoundTensionStrip_CQ4_4.modfem
|   |   |   |   CompoundTensionStrip_CQ4_7-0000.dat
|   |   |   |   compoundtensionstrip_cq4_7-0000.f04
|   |   |   |   compoundtensionstrip_cq4_7-0000.f06
```

```
│   │   │   │       compoundtensionstrip_cq4_7-0000.log
│   │   │   │       compoundtensionstrip_cq4_7-0000.mon1
│   │   │   │       compoundtensionstrip_cq4_7-0000.xdb
│   │   │   │       CompoundTensionStrip_CQ4_7.modfem
│   │   │   │
│   │   │   └───CQ8
│   │   │           CompoundTensionStrip_CQ8_10.dat
│   │   │           CompoundTensionStrip_CQ8_10.modfem
│   │   │           CompoundTensionStrip_CQ8_13.dat
│   │   │           CompoundTensionStrip_CQ8_13.modfem
│   │   │           CompoundTensionStrip_CQ8_16.dat
│   │   │           CompoundTensionStrip_CQ8_16.modfem
│   │   │           CompoundTensionStrip_CQ8_18.dat
│   │   │           CompoundTensionStrip_CQ8_18.modfem
│   │   │           CompoundTensionStrip_CQ8_22.dat
│   │   │           CompoundTensionStrip_CQ8_22.modfem
│   │   │           CompoundTensionStrip_CQ8_22_Refined.dat
│   │   │           CompoundTensionStrip_CQ8_22_Refined.modfem
│   │   │           CompoundTensionStrip_CQ8_28.dat
│   │   │           CompoundTensionStrip_CQ8_28.modfem
│   │   │           CompoundTensionStrip_CQ8_4.dat
│   │   │           CompoundTensionStrip_CQ8_4.modfem
│   │   │           CompoundTensionStrip_CQ8_7.dat
│   │   │           CompoundTensionStrip_CQ8_7.modfem
│   │   │
│   │   └───TensionPlate
│   │       └───PyDFEM
│   │               TensionPlate_PyDFEM.dat
```

```
│   │           TensionPlate_PyDFEM.modfem
│   │           TensionPlate_PyDFEM_Refined.dat
│   │           TensionPlate_PyDFEM_Refined.modfem
│   │
│   └───HoleInPlate
│           HoleInPlate-0002.dat
│           holeinplate-0002.f04
│           holeinplate-0002.f06
│           holeinplate-0002.log
│           holeinplate-0002.mon1
│           holeinplate-0002.xdb
│           HoleInPlate.modfem
│
├───Results
│   ├───!CompoundTensionStripDFEM_CQ4_V2
│   │   │   Convergence_CQ4.xlsx
│   │   │
│   │   ├───CompoundTensionStrip_CQ4_10-0000
│   │   │       01_gloabal_displacements.csv
│   │   │       02_gloabal_strains.csv
│   │   │       03_gloabal_stresses.csv
│   │   │
│   │   ├───CompoundTensionStrip_CQ4_13-0000
│   │   │       01_gloabal_displacements.csv
│   │   │       02_gloabal_strains.csv
│   │   │       03_gloabal_stresses.csv
│   │   │
│   │   ├───CompoundTensionStrip_CQ4_16-0000
```

```
│    │    │      01_gloabal_displacements.csv
│    │    │      02_gloabal_strains.csv
│    │    │      03_gloabal_stresses.csv
│    │    │
│    │    ├────────CompoundTensionStrip_CQ4_18-0000
│    │    │      01_gloabal_displacements.csv
│    │    │      02_gloabal_strains.csv
│    │    │      03_gloabal_stresses.csv
│    │    │
│    │    ├────────CompoundTensionStrip_CQ4_22-0000
│    │    │      01_gloabal_displacements.csv
│    │    │      02_gloabal_strains.csv
│    │    │      03_gloabal_stresses.csv
│    │    │
│    │    ├────────CompoundTensionStrip_CQ4_28-0000
│    │    │      01_gloabal_displacements.csv
│    │    │      02_gloabal_strains.csv
│    │    │      03_gloabal_stresses.csv
│    │    │
│    │    ├────────CompoundTensionStrip_CQ4_4-0000
│    │    │      01_gloabal_displacements.csv
│    │    │      02_gloabal_strains.csv
│    │    │      03_gloabal_stresses.csv
│    │    │
│    │    └────────CompoundTensionStrip_CQ4_7-0000
│    │           01_gloabal_displacements.csv
│    │           02_gloabal_strains.csv
│    │           03_gloabal_stresses.csv
```

```
│   │
│   ├───────!CompoundTensionStripDFEM_CQ8_V2
│   │   │   Convergence.xlsx
│   │   │
│   │   ├───────CompoundTensionStrip_CQ8_10
│   │   │       01_gloabal_displacements.csv
│   │   │       02_gloabal_strains.csv
│   │   │       03_gloabal_stresses.csv
│   │   │
│   │   ├───────CompoundTensionStrip_CQ8_13
│   │   │       01_gloabal_displacements.csv
│   │   │       02_gloabal_strains.csv
│   │   │       03_gloabal_stresses.csv
│   │   │
│   │   ├───────CompoundTensionStrip_CQ8_16
│   │   │       01_gloabal_displacements.csv
│   │   │       02_gloabal_strains.csv
│   │   │       03_gloabal_stresses.csv
│   │   │
│   │   ├───────CompoundTensionStrip_CQ8_18
│   │   │       01_gloabal_displacements.csv
│   │   │       02_gloabal_strains.csv
│   │   │       03_gloabal_stresses.csv
│   │   │
│   │   ├───────CompoundTensionStrip_CQ8_22
│   │   │       01_gloabal_displacements.csv
│   │   │       02_gloabal_strains.csv
│   │   │       03_gloabal_stresses.csv
```

```
│   │   │
│   │   ├────CompoundTensionStrip_CQ8_28
│   │   │      01_gloabal_displacements.csv
│   │   │      02_gloabal_strains.csv
│   │   │      03_gloabal_stresses.csv
│   │   │
│   │   ├────CompoundTensionStrip_CQ8_4
│   │   │      01_gloabal_displacements.csv
│   │   │      02_gloabal_strains.csv
│   │   │      03_gloabal_stresses.csv
│   │   │
│   │   └────CompoundTensionStrip_CQ8_7
│   │          01_gloabal_displacements.csv
│   │          02_gloabal_strains.csv
│   │          03_gloabal_stresses.csv
│   │
│   └────!OpenHolePlate
│       ├────HoleInPlate-0002
│       │      01_gloabal_displacements.csv
│       │      02_gloabal_strains.csv
│       │      03_gloabal_stresses.csv
│       │
│       └────TensionPlate_PyDFEM
│              01_gloabal_displacements.csv
│              02_gloabal_strains.csv
│              03_gloabal_stresses.csv
│
└────Sympy_ShapeFunc_Derivation
```

B_Matrix.ipynb

CQUAD4_SympyCalc.html

CQUAD4_SympyCalc.ipynb

CQUAD6_SympyCalc.html

CQUAD6_SympyCalc.ipynb

CQUAD7_SympyCalc.html

CQUAD7_SympyCalc.ipynb

CQUAD8_SympyCalc.html

CQUAD8_SympyCalc.ipynb

CTRIA3_SympyCalc.ipynb